BRAD BURCH

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SUMMARY

Senior full-stack software engineer with 9 years building production systems at Salesforce, Total Brain, Front, and Pawservation. Architected and own a booking, payments, and Claude agent platform (MCP, OAuth 2.1 + PKCE, Stripe Connect) on serverless infrastructure that moved more than half of bookings to self-serve, from data model through evals, on-call, and incident response.

SKILLS

Languages & Frameworks: TypeScript, Python, JavaScript, SQL, Java, Scala, React 19, Node.js, Django, Hono, Spring, Play
Infrastructure: PostgreSQL, SQLite, AWS (DynamoDB, SQS), serverless functions, message queues and workflow orchestration (Cloudflare Workers), S3-compatible storage, MSSQL, Docker, Kubernetes, REST APIs, webhooks, distributed systems
AI / LLM: Claude (Haiku, Sonnet), AI agents, Vercel AI SDK, Model Context Protocol (MCP), tool calling, LLM evals, prompt caching, prompt-injection defense, AI cost metering, Claude Code, Cursor
Payments & Security: Stripe (Connect, Billing, Checkout), OAuth 2.1 + PKCE, JWT, SAML 2.0, HMAC webhook verification, AES-GCM, SSRF prevention, multi-tenant isolation
Testing & Delivery: Vitest, Playwright, contract tests, GitHub Actions, Jenkins, CI/CD, observability (structured logging, alerting)

EXPERIENCE

Software Engineer 2022 - Present

Pawservation, San Francisco, CA (pawservation.com) | booking, payments, and AI platform

- Built and operate a production serverless TypeScript platform (React, REST APIs, SQLite) handling 60+ bookings/month and 821 imported client payments, replacing 5 to 15 hours/month of manual operations.
- Moved over half of bookings to self-serve with an LLM booking agent (Claude) on web chat, WhatsApp, and MCP, graded on 5,418 eval turns with 0 unapproved writes.
- Raised the AI booking agent's pass rate from 85% to 93% across 1,251 eval cases by triaging 292 failures into 22 root causes and moving date parsing into deterministic code.
- Traced CPU-limit kills to a logo re-decoded on every invoice, dropping PDF CPU time 88% (154 ms to 18 ms) and recovering 4 months of failed invoice runs.
- Lowered per-turn model cost about 8x by moving chat and the sitter assistant to a cheaper model tier.
- Stopped under-billing 92% of overnight stays (109 of 118 charged as one night) by defaulting the invoicer to per-night billing.
- Eliminated the source of 97% of the account's daily database writes by making a 15-minute calendar sync skip unchanged rows, before a 100,000/day cap applied.
- Guarded against double-booking on 4 channels with a database lease per booking write and a re-check at approval.
- Consolidated 4 cron triggers into 1 hourly trigger running durable workflows keyed by the hour, so a double-fired cron runs once and never emails invoices twice.
- Added auth-failure, stuck-job, and invoice-completion alerting after 3 silent integration failures each ran for weeks undetected.
- Shipped an MCP server behind a hand-built OAuth 2.1 + PKCE authorization server, adding a third AI booking channel clients use from their own AI assistant.
- Isolated tenant config, pricing, capacity, and bookings by routing every SQL query through one business-scoped module, with 0 cross-tenant writes across 5,418 eval turns.

Software Engineer 2022

Front, San Francisco, CA (role eliminated in company-wide layoff)

- Delivered a Node.js REST API for account-level email rules for multi-domain enterprise customers, replacing per-inbox setup.
- Upgraded security in the core JavaScript email library handling 200K+ outbound messages/month.

Software Engineer 2019 - 2021

Total Brain, San Francisco, CA

- Shortened enterprise onboarding from 2+ weeks to ~3 days by standardizing SAML 2.0 and JWT SSO, HubSpot, and Box integrations as engineering point of contact on Sales and CS calls.
- Halved average API response times by migrating MSSQL to DynamoDB with a redesigned schema; mentored junior engineers.
- Contributed to a 60% lift in new sign-ups by shipping a Scala backend integration for Apple, Google, and Facebook sign-in.

Software Engineer 2017 - 2018

Salesforce, San Francisco, CA

- Led delivery of a config builder and validator used by internal teams, scoping milestones and coordinating partner teams.
- Trimmed internal test setup time 40% with a React management UI, and wrote an OAuth2 security filter for microservice access.

PROJECTS

- **MCP Auth Kit (open source):** OAuth 2.1 + PKCE kit for MCP servers with rate limits, scoped tools, and two-phase confirmation.
- **Roadrunner (open source):** Python/Django and PostgreSQL app matching eBird and iNaturalist sightings to Strava activities via OAuth, webhooks, and a deferred re-check queue.

EDUCATION

DePauw University, Bachelor of Arts in Computer Science 2017